Time Series Analysis and Forecasting of Undertrial Prison Population in India

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ABSTRACT

This study presents a time series analysis of the undertrial prison population in India using annual data from 1995 to 2023. Undertrial prisoners, who are individuals awaiting completion of judicial proceedings, serve as a critical indicator of judicial efficiency and systemic backlog. Using the Box–Jenkins methodology, the series is examined for stationarity, transformed where necessary, and modelled using an ARIMA framework with drift. Diagnostic checks are performed to assess model adequacy, and short-term forecasts with confidence intervals are generated. The study provides conditional projections based on historical institutional patterns and highlights that meaningful departures from projected trends would require structural changes beyond those observed in the past.

INTRODUCTION

The undertrial prison population has emerged as a major concern within India's criminal justice system. Undertrials are individuals who have been charged with offences but have not yet been convicted. Prolonged detention of undertrials raises serious issues related to due process, prison overcrowding, and access to justice. Unlike total prison population figures, which combine convicts and other categories, undertrial counts directly reflect delays in investigation, trial, and case disposal.

While descriptive statistics show a steady increase in undertrial numbers over time, visual inspection alone cannot determine whether this growth represents a persistent structural process or random variation. Nor can it provide reliable insight into future trajectories under existing institutional conditions. This project addresses these limitations by applying formal time series techniques to model the undertrial population as a stochastic process evolving over time.

The objective of this study is not to evaluate specific judicial reforms, but to understand long-run behaviour, separate structural trends from short-term fluctuations, and produce uncertainty-aware forecasts based on historical patterns.

DATASET DESCRIPTION

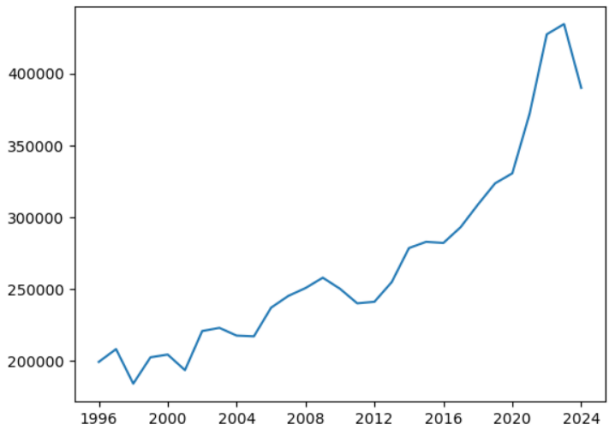
The dataset consists of annual observations on the total undertrial prison population in India from 1995 to 2023. The data is sourced from official publications of the National Crime Records Bureau (NCRB), specifically the Prison Statistics India reports. Only years with consistent and comparable reporting standards were included to ensure reliability.

The variable of interest is the total number of undertrial prisoners at the national level, measured at the end of each calendar year. The annual frequency is appropriate given the institutional nature of judicial processes and the availability of official data.

DATA PREPROCESSING

The raw data was imported using Python and processed to ensure compatibility with time series methods. The year variable was converted into a datetime format corresponding to year-end observations and set as the time index. The series was explicitly defined at an annual frequency. No artificial smoothing or interpolation was applied, preserving the integrity of the original observations.

EXPLORATORY DATA ANALYSIS



The time plot shows a clear long-term upward trend in the undertrial prison population. Although some years display temporary declines or slower growth, these fluctuations are short-lived and do not change the overall increasing pattern. No seasonal or cyclical behaviour is evident, which is expected for annual data. The strong trend suggests the need for formal stationarity testing and time series modelling.

AUGMENTED DICKEY-FULLER (ADF) TEST FOR STATIONARITY

Purpose of the test: The Augmented Dickey–Fuller (ADF) test is used to check whether a time series is stationary or not. In time series analysis, stationarity is important because many models require the series to have stable behaviour over time. The ADF test helps determine whether the

series shows a persistent trend or whether it fluctuates around a constant level. Based on the test result, it can be decided whether differencing is required before fitting a time series model.

Hypothesis:

Null Hypothesis (H_0): The time series is non-stationary i.e. has a unit root

Alternative Hypothesis (H_1): The time series is stationary i.e. does not have a unit root

Decision Rule:

If the p-value ≤ 0.05, reject Null Hypothesis and conclude that the series is stationary.

ADF Test Results:

```
# checking if data is stationary or not
from statsmodels.tsa.stattools import adfuller
adftest = adfuller(data['Undertrials'])
def adf_test():
    pvalue = adftest[1]
    print('pvalue of adfuller test is:', pvalue)
    if pvalue <= 0.05:
        print("Stationary dataset")
    else:
        print("Non-stationary dataset")
adf_test()
```

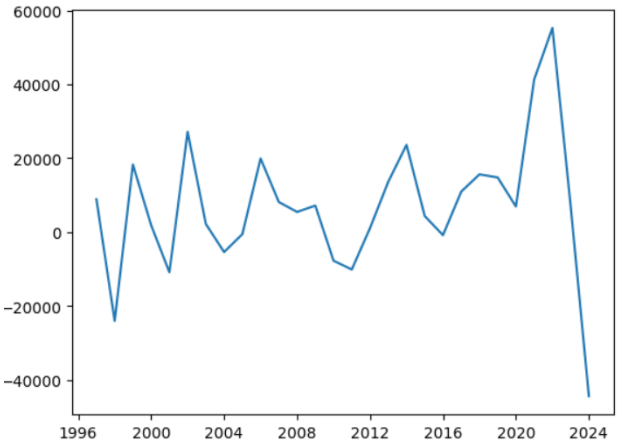
```
pvalue of adfuller test is: 0.9932192502658994
Non-stationary dataset
```

The test resulted in a p-value greater than 0.05. Null hypothesis is accepted and hence the series is non-stationary in its level form. To achieve stationarity and make the data suitable for time series modelling, first-order differencing is applied in the next step.

```
# implementing differencing
data['Undertrials_diff'] = data['Undertrials'].diff()

plt.plot(data['Undertrials_diff'])
plt.show()
```

Below is the time plot of the first-order differenced undertrial prison population series.



After applying first-order differencing, the Augmented Dickey–Fuller (ADF) test was again implemented on the transformed series.

```
adftest = adfuller(data['Undertrials_diff'].dropna())
adf_test()
```

```
pvalue of adfuller test is: 0.0001948087199205976
Stationary dataset
```

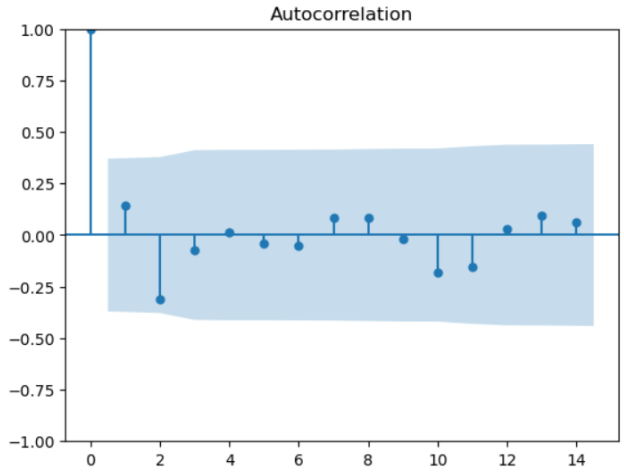
The test yielded a p-value less than 0.05. Hence, the null hypothesis is rejected, which indicates that the differenced undertrial prison population series is stationary. Now, since stationarity is achieved, the series is suitable for time series modelling.

MODEL SELECTION RATIONALE

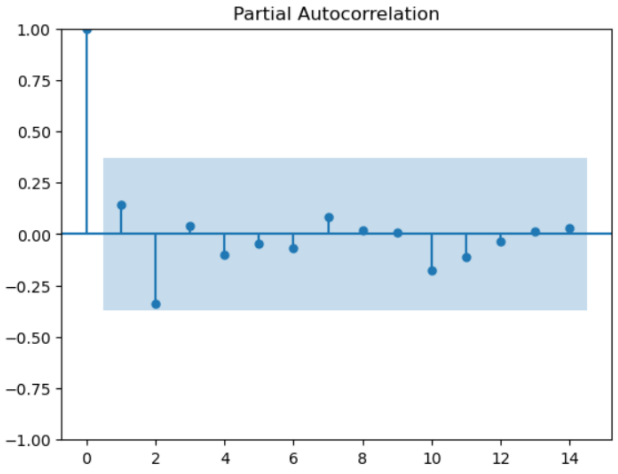
An ARIMA model is chosen because the series is non-stationary in its level form and becomes stationary only after first-order differencing. Since pure AR and MA models require stationarity without differencing, they are not suitable in this case. ARIMA allows the inclusion of differencing along with autoregressive and moving average components, making it appropriate for the given data.

IDENTIFICATION OF ARIMA PARAMETERS USING ACF AND PACF

After selecting the ARIMA framework, the next step is to identify the appropriate values of the autoregressive (p) and moving average (q) parameters. This is done using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots of the stationary (first-differenced) series. The PACF plot is used to identify p while the ACF plot is used to identify q. These plots provide insights into the lag structure of the data.



From the ACF plot of the differenced series, only the first lag shows noticeable significance, while the remaining lags lie within the confidence limits. This suggests that the dependence on past error terms is limited, and hence a moving average component of order one ($q = 1$) is sufficient.



Unlike the ACF, the PACF measures the direct relationship between the series and its lagged values by removing the effect of intermediate lags. From the PACF plot of the first-differenced series, a significant spike is observed at lag one, while the remaining lags lie within the confidence bounds. This suggests that the current value of the series

is mainly influenced by its immediately preceding value, and hence an autoregressive component of order one ($p = 1$) is appropriate.

Based on the ACF and PACF analysis of the stationary series, an **ARIMA(1,1,1)** model is selected for further analysis. This model captures both the autoregressive and moving average behaviour observed in the differenced undertrial prison population series.

ARIMA(1,1,1) MODEL ESTIMATION

```
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(data['Undertrials'], order=(1,1,1),trend='t')
result = model.fit()
print(result.summary())
```

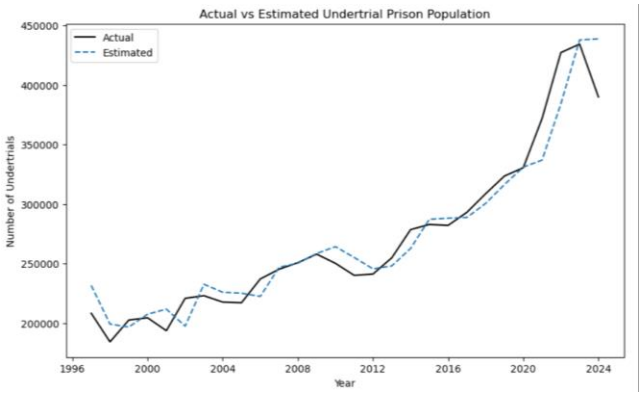
```
                               SARIMAX Results
==============================================================================
Dep. Variable:           Undertrials   No. Observations:                   29
Model:                 ARIMA(1, 1, 1)   Log Likelihood                -313.205
Date:                Mon, 15 Dec 2025   AIC                            634.410
Time:                        19:59:38   BIC                            639.739
Sample:                  12-31-1995   HQIC                           636.039
                       - 12-31-2023
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
x1          6805.3928   3678.426      1.850      0.064    -404.189     1.4e+04
ar.L1         -0.4922      0.495     -0.995      0.320      -1.462       0.477
ma.L1          0.6440      0.494      1.302      0.193      -0.325       1.613
sigma2      2.978e+08      0.007   4.08e+10      0.000    2.98e+08    2.98e+08
===================================================================================
Ljung-Box (L1) (Q):                   1.07   Jarque-Bera (JB):                 2.61
Prob(Q):                              0.30   Prob(JB):                         0.27
Heteroskedasticity (H):               2.96   Skew:                             0.10
Prob(H) (two-sided):                  0.12   Kurtosis:                         4.48
===================================================================================
```

The ARIMA(1,1,1) model was estimated for the undertrial prison population series to capture its dynamic behaviour over time. The model incorporates both autoregressive and moving average components, along with a deterministic trend, to reflect the persistent upward movement observed in the data.

The diagnostic results suggest that the fitted model adequately explains the time dependence in the series. The Ljung–Box test indicates no significant autocorrelation remaining in the residuals, implying that the model has successfully captured the underlying structure of the data. Furthermore, the Jarque–Bera test does not show strong evidence against normality of residuals. Overall, the ARIMA(1,1,1) model provides a reasonable fit and is suitable for producing in-sample estimates and conditional forecasts based on historical patterns.

ACTUAL VS ESTIMATED VALUES

After estimating the ARIMA model, the fitted (estimated) values are compared with the actual undertrial prison population to assess the in-sample performance of the model.

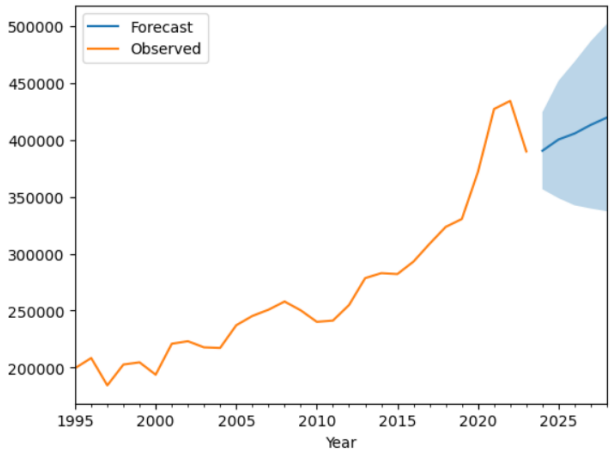


The estimated values closely follow the overall movement of the observed series, capturing the long-term trend as well as short-term variations to a reasonable extent.

This comparison indicates that the model provides an adequate representation of the historical data and is able to explain the underlying structure of the series. The good alignment between actual and estimated

values supports the suitability of the fitted model for further forecasting.

FUTURE FORECASTS



The fitted time series model is used to generate forecasts for the undertrial prison population for a five year horizon.

These forecasts are obtained by extending the historical pattern observed in the data and represent expected future values under the assumption that past behaviour continues.

The forecasted values appear smooth because the model focuses on the overall direction of the series rather than predicting unpredictable year-to-year changes. In annual data, such short-term fluctuations do not follow a stable pattern and cannot be reliably forecasted.

The confidence intervals widen as we move further into the future. This indicates that uncertainty increases with time, meaning the forecasts become less precise for longer horizons. This also implies that future outcomes are sensitive to changes in how the system operates.

Since the forecasts are generated using past data, any major deviation from the projected path would require a change in the underlying system. In other words, if future undertrial numbers move significantly away from these forecasts, it would suggest that institutional functioning has changed in a way not reflected in historical data.

LIMITATIONS OF THE STUDY

The analysis relies exclusively on historical undertrial prison population data and does not account for external influences such as policy changes, judicial reforms, or administrative interventions. As a result, the forecasts should be interpreted as conditional projections based on past patterns rather than exact predictions. Any substantial change in institutional functioning may therefore lead to outcomes that differ from the projected results.

CONCLUSION

This project examined the undertrial prison population in India through time series analysis to understand how the system behaves over time rather than relying only on year-to-year changes. By formally testing the data and building a suitable time series model, the study captured the underlying pattern in the growth of the undertrial population and used this structure to generate future projections.

Although there are periods of decline in undertrial numbers, these declines are not sustained. This shows that short-term reductions cannot be treated as real improvement. True and lasting reduction in the undertrial population would require

structural reforms in judicial processing, rather than temporary changes driven by yearly variation. The forecasts produced in this study provide a conditional picture of the future, showing what may happen if the system continues to function as it has in the past.

Ultimately, this project highlights that behind every number in the dataset is a human being waiting for justice. By using time series modelling, the study converts raw data into insights that can support informed discussion, planning, and reform. The work demonstrates how statistical tools can be used not only to analyse trends, but also to reflect on the functioning of important public institutions.